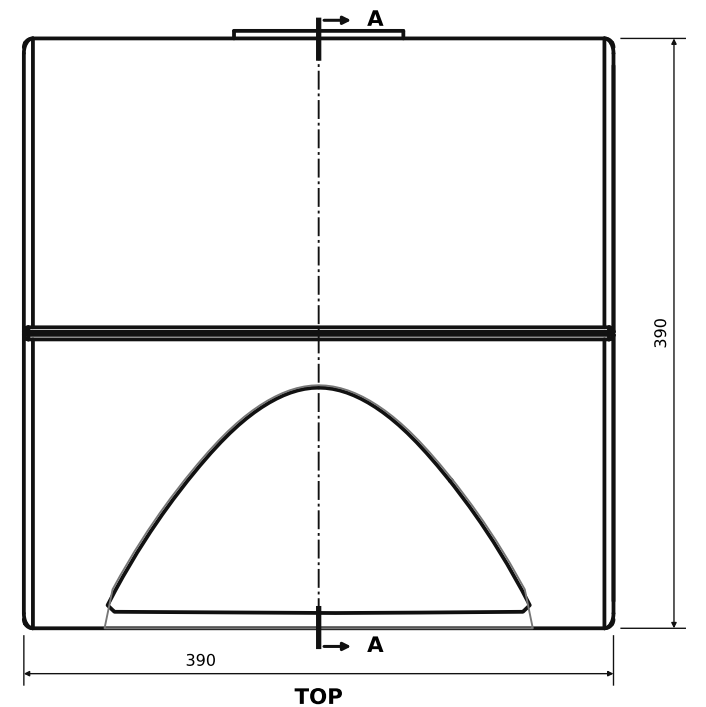
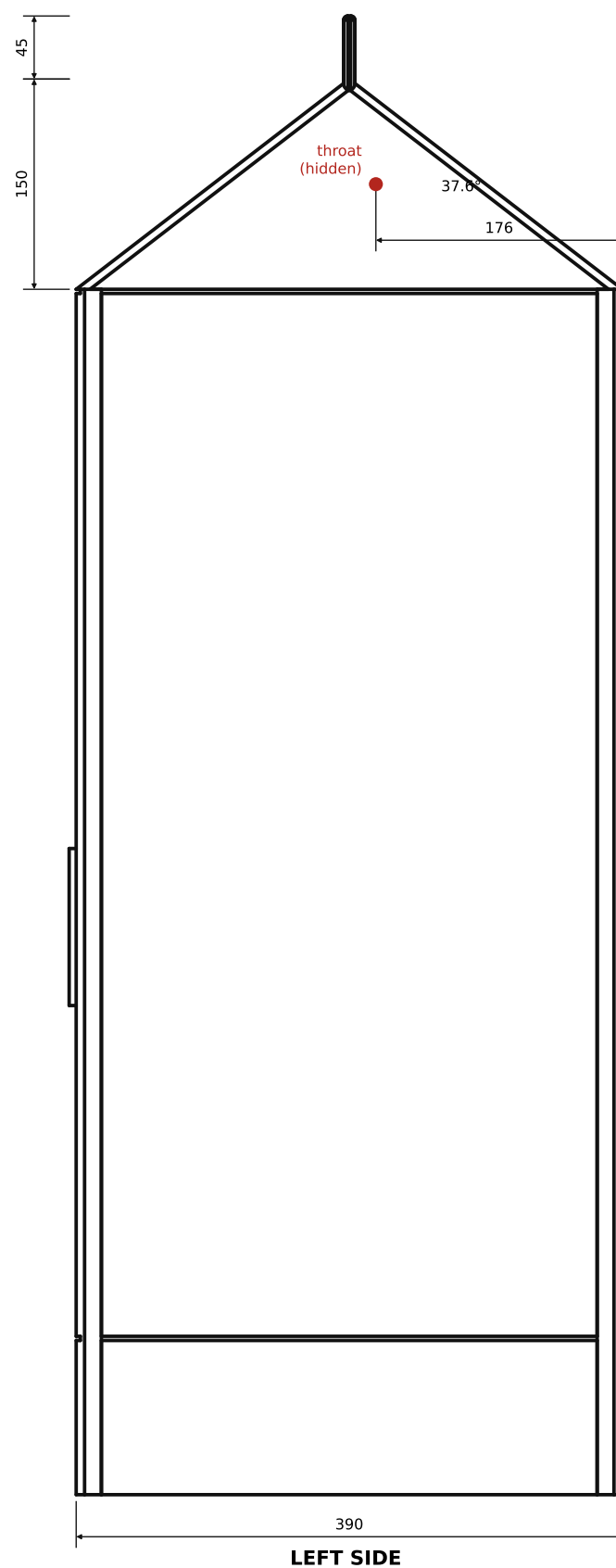
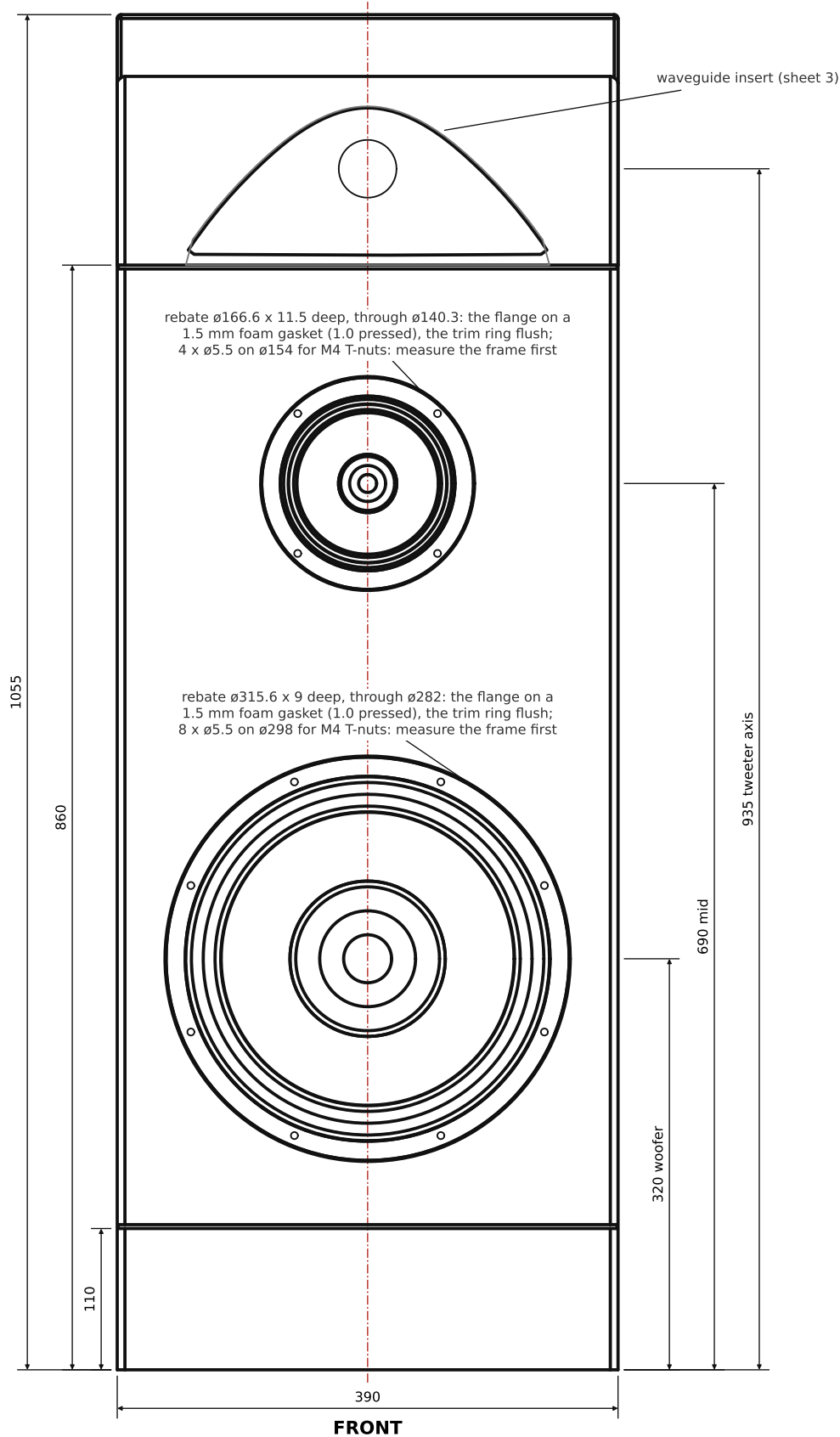




black: the waveguide's mouth, where its lip meets the roof
grey: the insert's seam

Notes

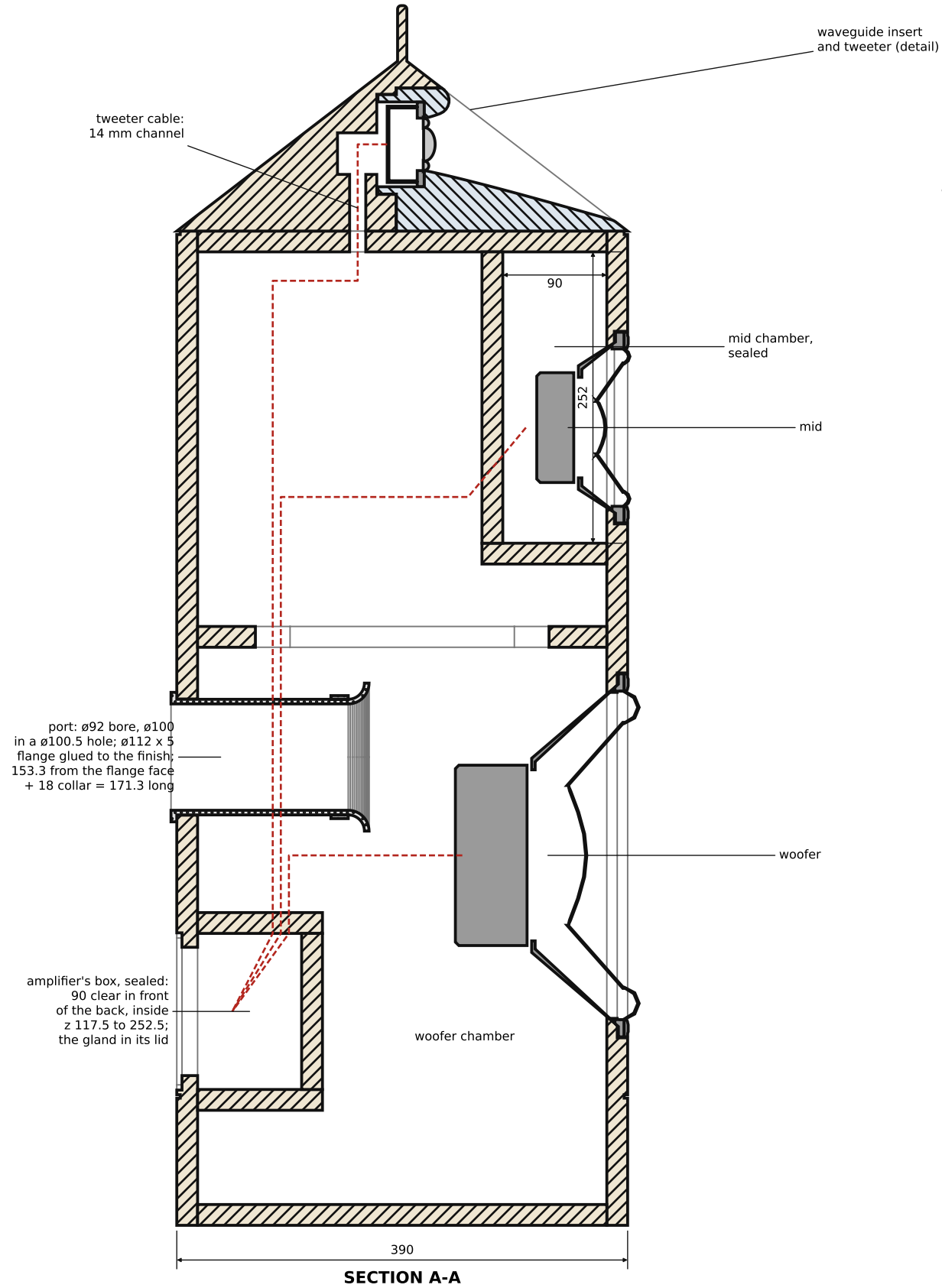
1. Cabinet: 18 mm Baltic birch, glued. The vertical corners and the gable's hips are rounded 6 mm, the fin's edges 3 mm. 3 x 3 mm shadow lines at z 110 and z 857.
2. The drivers sit flush: each frame on a 1.5 mm closed-cell foam gasket in its rebate, M4 button heads into T-nuts, and a 3 mm printed trim ring over the flange (stl/trim-ring-*.stl: a channel in its back clears the screw heads), held by three 5 mm dots of neutral-cure silicone, 0.8 reveal all round.
3. The tweeter is at the throat of a waveguide insert in the roof (sheet 3): axis z 935, 176 behind the front face. The shape is set by simulation (fab/out/acoustics/waveguide).
4. The back (port, amplifier, label) is drawn on sheet 4 with the wiring.
5. Finish: colour coat and 2K clear; the Facts printed between them.

earmilk floorstander | General arrangement

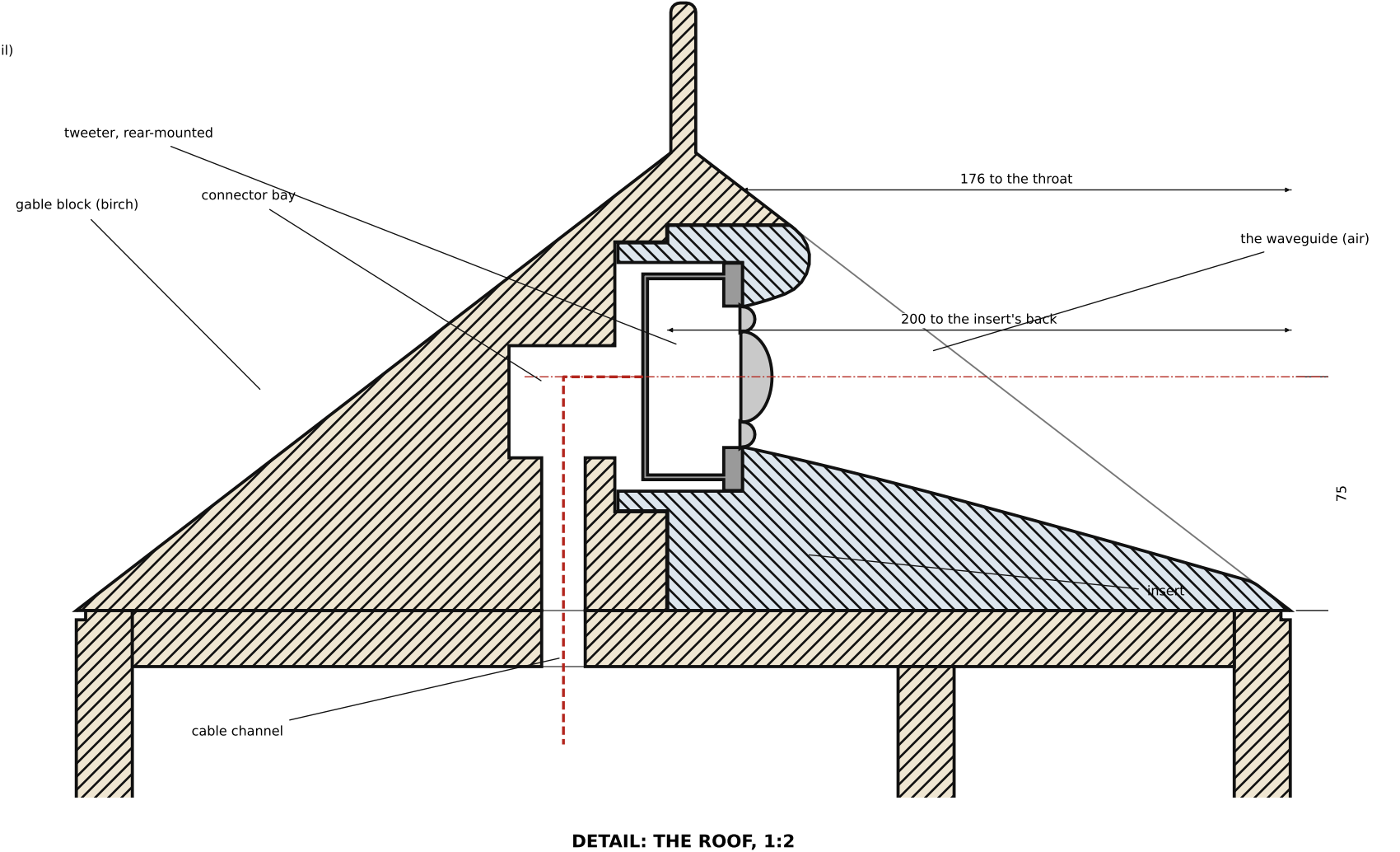
Front, left side and top at 1:5. Heights from the floor; depths from the front face.

mm | issue 2026-10-09 | from fab/sheets.py (the CAD) | fits as dimensioned

Tolerances: panels ±0.5, pocket depths ±0.2, holes +0.2/-0, prints ±0.15. Check bought parts first.



red dashed: the cables' runs to the amplifier (sheet 4)



The roof

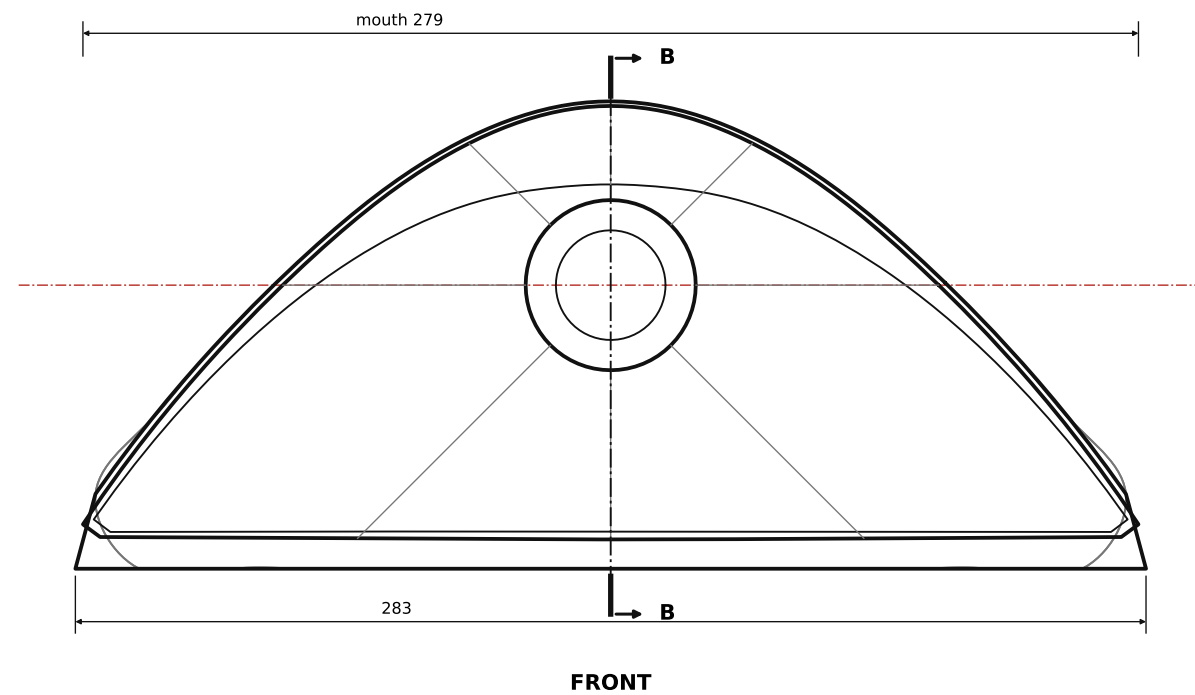
1. The waveguide insert (light blue) sits in a pocket in the gable block, 0.3 mm clear all round. It slides out forward, level, with the tweeter on it.
2. The tweeter is rear-mounted: it goes in from behind through a $\varnothing 73.4$ bore and its flange seats on the ring round the throat, its dome and surround filling the 45 mm throat, so the wall runs on from the surround with no step.
3. Its lead leaves the boss's open back into the connector bay behind it ($\varnothing 36 \times 35$ deep), where it plugs into the cabinet's lead; that runs down a 14 mm channel through the block and the top panel, sealed by a grommet under the top panel.
4. Four magnets and two pins in the insert's back face meet their partners in the pocket's back wall (sheet 3).

earmilk floorstander | Section on the centreline

Section A-A on the centre plane (x 195) seen from the left at 1:5, and the roof at 1:2. Cut wood hatched; the insert hatched the other way.

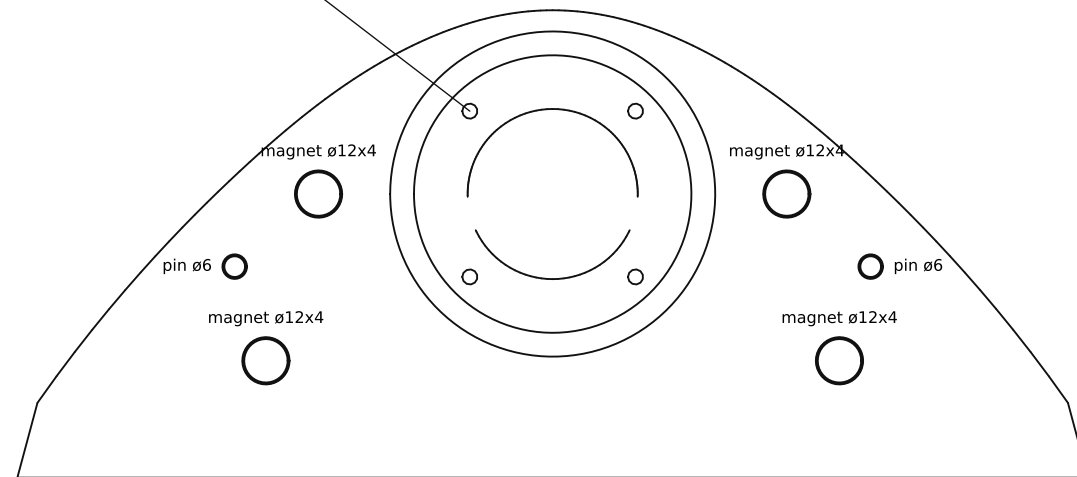
mm | issue 2026-10-09 | from fab/sheets.py (the CAD) | fits as dimensioned

Tolerances: panels ± 0.5 , pocket depths ± 0.2 , holes $+0.2/-0$, prints ± 0.15 . Check bought parts first.

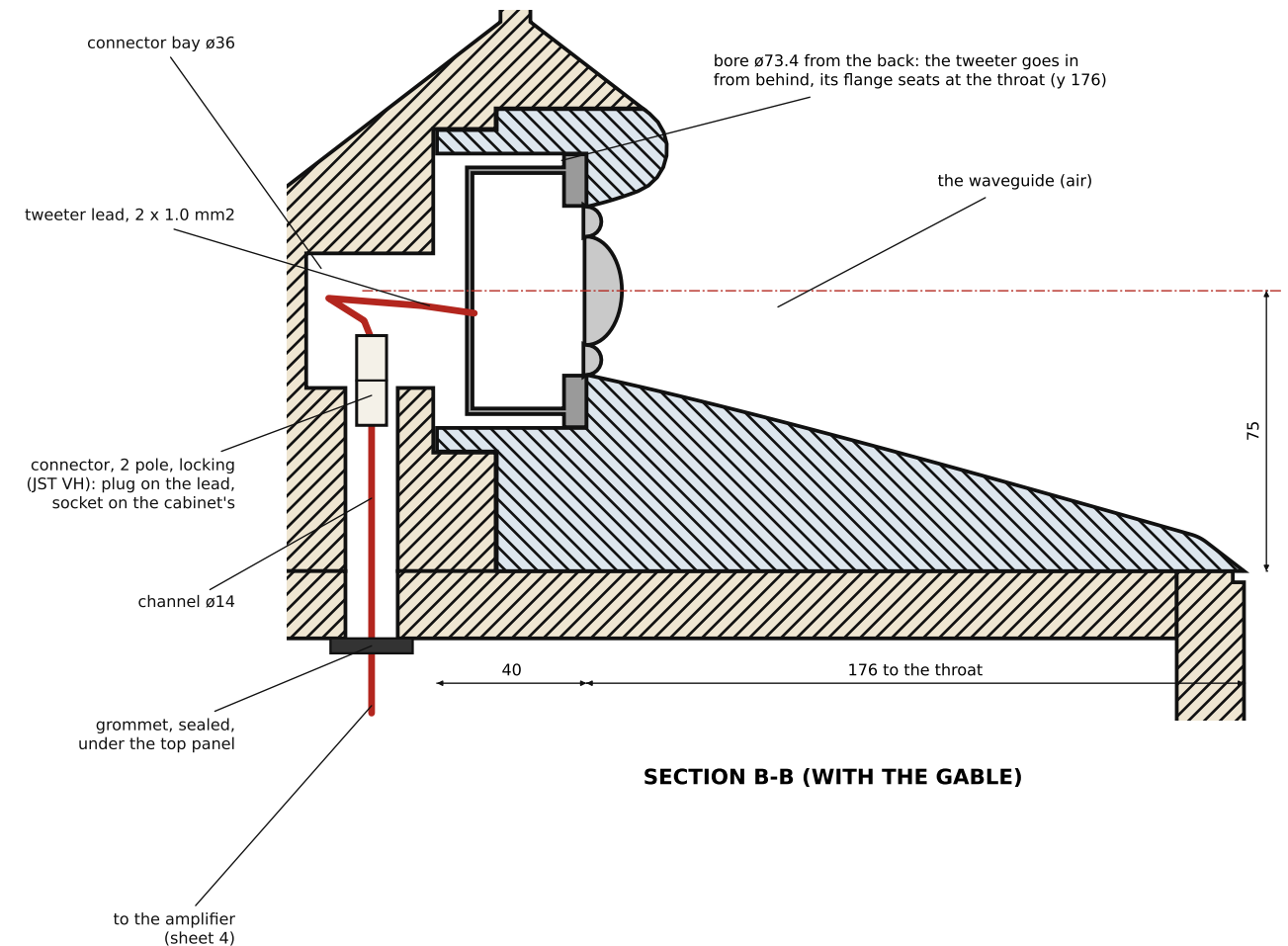


FRONT

4 x $\emptyset 4 \times 6$ on $\emptyset 62$ in the seat, for M3 x 5.7 heat-set brass inserts: measure the tweeter first



BACK (THE FACE THAT MEETS THE POCKET)



SECTION B-B (WITH THE GABLE)

Fitting the tweeter and the insert

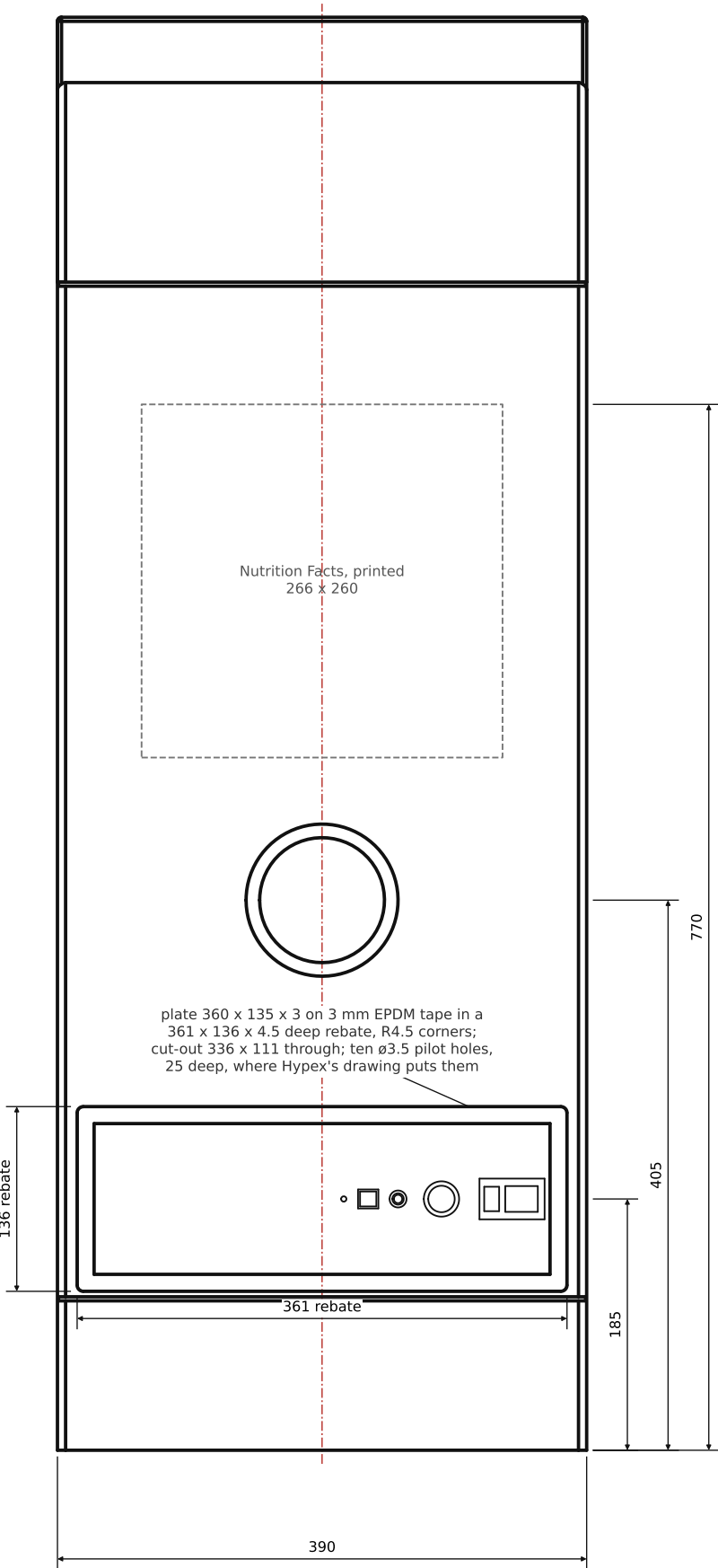
1. Solder the tweeter's lead (2 x 1.0 mm2, 120 mm) to its tabs and crimp the plug half of a two-pole locking connector (JST VH) on its end.
2. Melt 4 M3 x 5.7 heat-set brass inserts into the seat round the throat ($\emptyset 4 \times 6$ holes on $\emptyset 62$). Pass the tweeter in from behind through the $\emptyset 73.4$ bore, dome first, and seat its flange on a 0.5 mm foam gasket; screw it to the inserts.
3. Glue the magnets into the insert's back with epoxy, polarity marked, and their partners into the pocket's back wall, opposite poles out. Press the two pins into the insert.
4. The cabinet's lead ends in the socket 250 mm above the grommet. With the insert out, draw the socket forward out of the bay through the empty pocket, plug the tweeter's lead into it, and tuck both and the slack back into the bay.
5. Slide the insert in, level, until the pins seat and the magnets pull it home: its face flush with the roof, the seam even.
6. Service: lift it out with a suction lifter on its flat border (or a pull loop under its lower edge), unplug, and the tweeter comes out with it.
7. Fixings, from the centreline and the insert's base (mirrored in the pocket's wall): magnets ± 75.9 across, 30.9 up; ± 61.9 across, 75.0 up; pins ± 84.1 across, 55.8 up. Magnets $\emptyset 12 \times 4$ in $\emptyset 12.2 \times 4.3$ holes, epoxied, poles marked; pins $\emptyset 6 \times 10$ pressed into $\emptyset 5.9 \times 5.5$ in the insert, a slip fit in $\emptyset 6.2 \times 5.5$ in the wall.

earmilk floorstander | Waveguide insert and tweeter mount

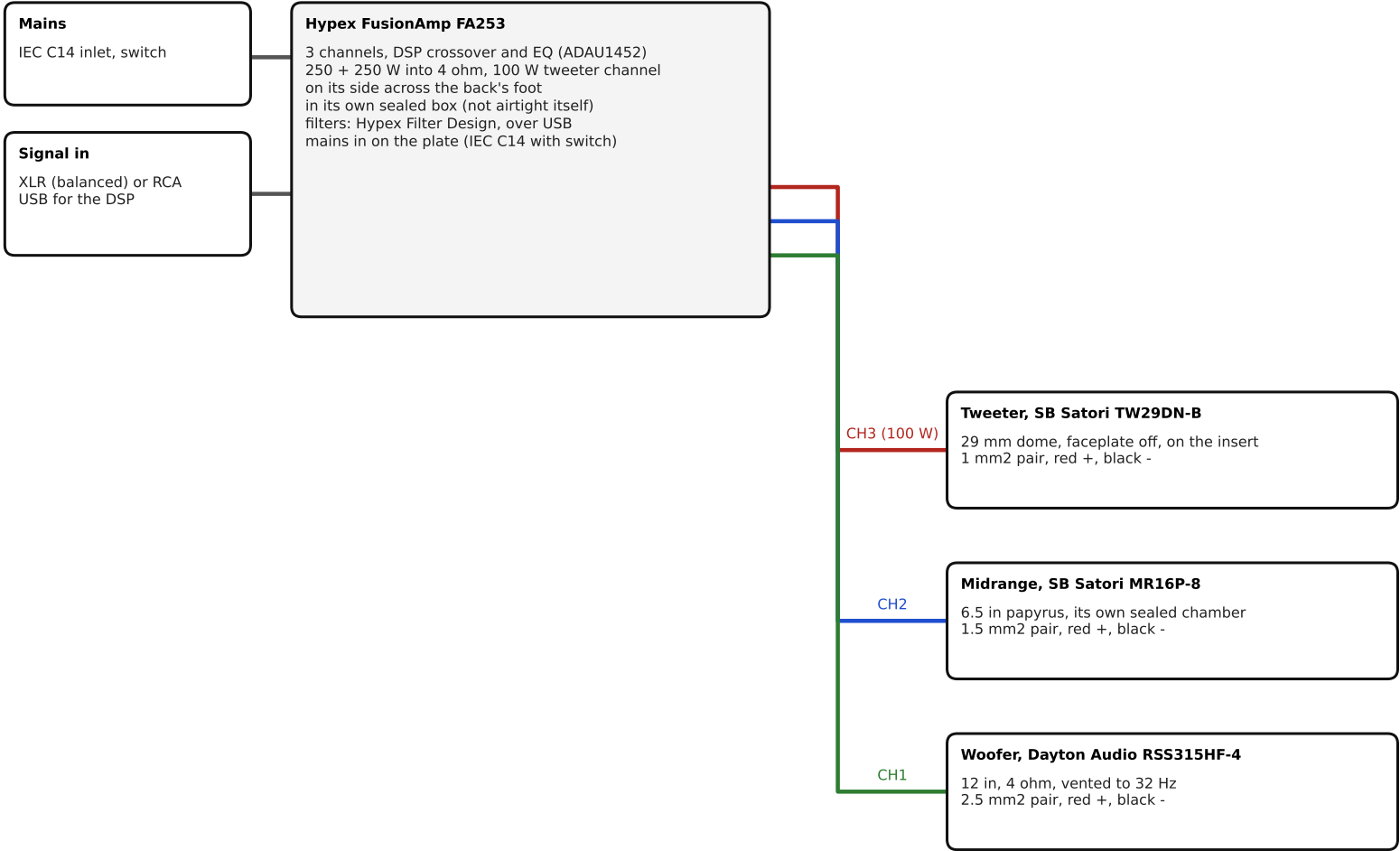
The insert at 1:2: front, section on the axis, back; the pocket; fitting and service.

mm | issue 2026-10-09 | from fab/sheets.py (the CAD) | fits as dimensioned

Tolerances: panels ± 0.5 , pocket depths ± 0.2 , holes $+0.2/-0$, prints ± 0.15 . Check bought parts first.



BACK



Runs, connectors and seals

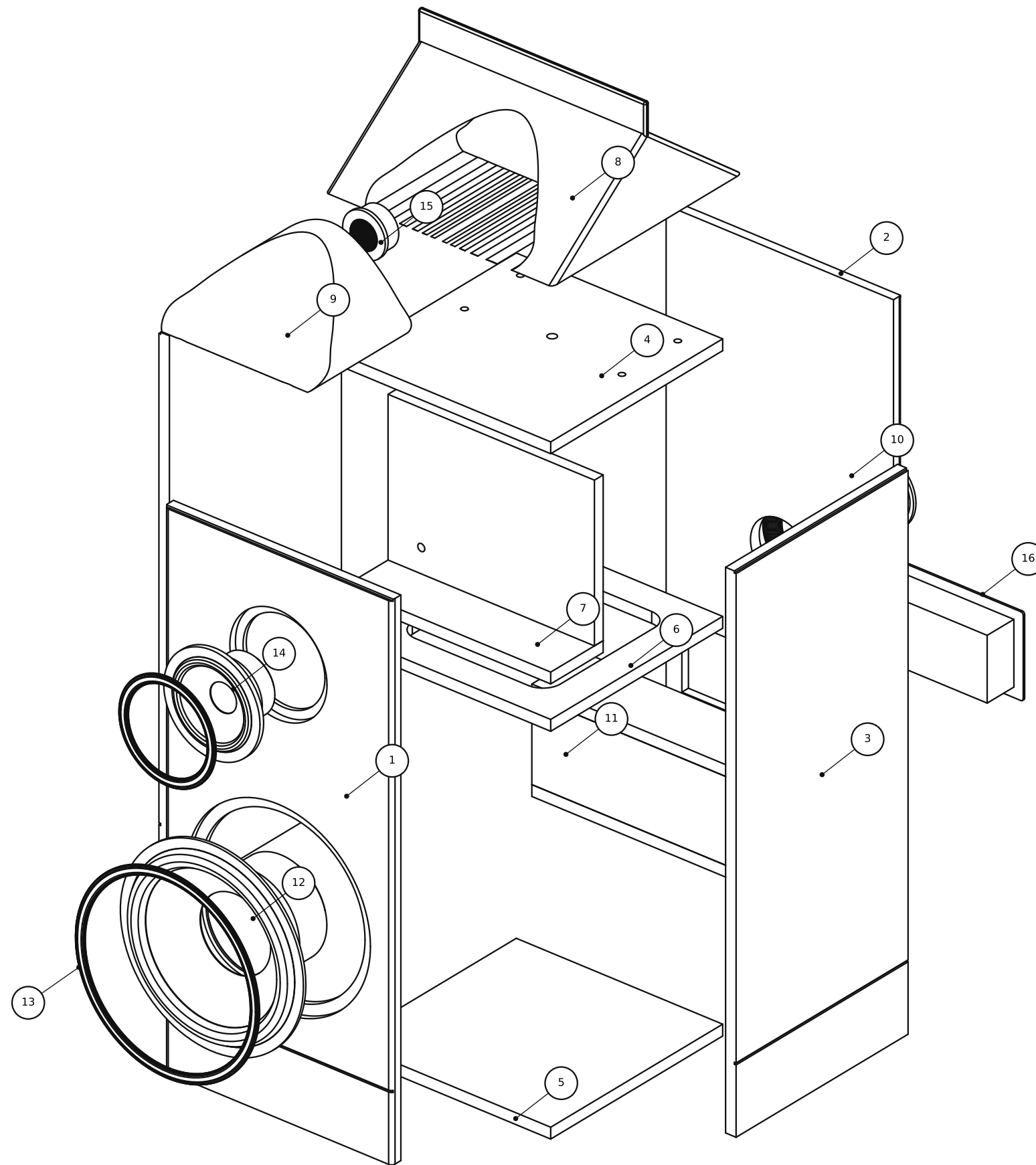
1. Woofer (CH1): from the amplifier box's gland straight to the woofer's terminals, 0.6 m of 2.5 mm2. Fit 6.3 mm push-on terminals to suit the driver.
2. Mid (CH2): from the gland up through the brace's window, 30 mm in from its back edge, then through the 12 mm hole in the mid chamber's divider (x 75, z 630), 1.0 m of 1.5 mm2. Seal the hole round the cable with putty: the mid's chamber must stay closed.
3. Tweeter (CH3): from the gland up through the brace's window to the top panel, through the 14 mm channel in the top panel and the gable block, 1.3 m of 1.0 mm2, ending in the socket of a two-pole locking connector (JST VH) in the bay behind the insert. The supplied 1.25 m Hypex harness is too short: extend it.
4. Seal the tweeter cable where it passes the top panel with a rubber grommet and sealant. The insert's pocket is open to the room through its 0.3 mm seam, so an unsealed channel would be a leak in the woofer's box.
5. The amplifier box's lid: an M25 x 1.5 long-thread (15 mm) gland with a three-hole seal for the three pairs, in a ø25.5 hole 314 from the left side's inner face and 43 from the lid's front edge, its lock nut in a ø40 x 8 counterbore under the lid; seal it. The box itself is glued and sealed; its front is the woofer chamber's wall.
6. Polarity, delay and the crossover (about 2.8 kHz tweeter to mid, from the waveguide study) are set in the DSP: wire red to + throughout and let the filters do the rest.

earmilk floorstander | Back and wiring

The back at 1:5 and the amplifier, drivers, cables and connectors. Active: the amplifier's DSP is the crossover.

mm | issue 2026-10-09 | from fab/sheets.py (the CAD) | fits as dimensioned

Tolerances: panels ±0.5, pocket depths ±0.2, holes +0.2/-0, prints ±0.15. Check bought parts first.



Parts

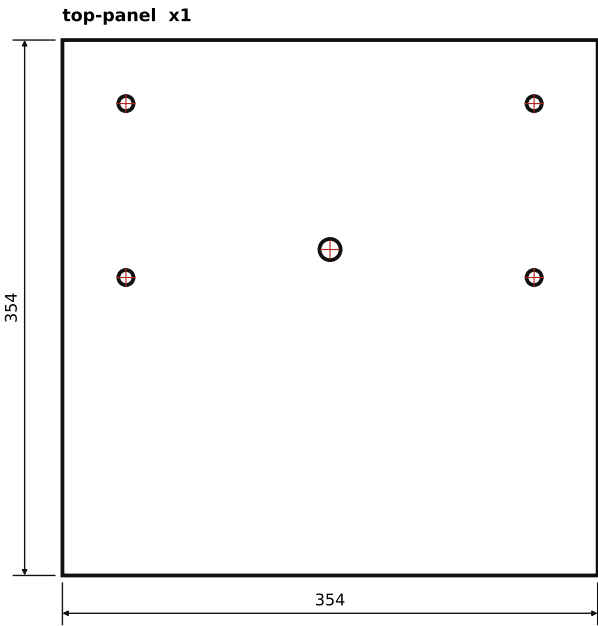
- ① **Front baffle x1**
18 mm Baltic birch, CNC: cutouts and rebates for the drivers
- ② **Back x1**
18 mm birch: port hole, the amplifier's cutout and rebate
- ③ **Sides x2**
18 mm birch
- ④ **Top panel x1**
18 mm birch, the tweeter cable's 14 mm hole, four dowel holes
- ⑤ **Bottom panel x1**
18 mm birch
- ⑥ **Window brace x1**
18 mm birch
- ⑦ **Mid chamber shelf and divider x2**
18 mm birch; the divider's 12 mm cable hole
- ⑧ **Gable block x1**
laminated birch, CNC: the insert's pocket and the cable channel
- ⑨ **Waveguide insert x1**
CNC from solid (MDF, HDPE) or printed (SLA, MJF); magnets, pins
- ⑩ **Port x1**
printed tube and flare collar, 92 bore
- ⑪ **Amplifier box x3**
18 mm birch floor, lid (gland) and front, glued and sealed
- ⑫ **Woofer x1**
Dayton Audio RSS315HF-4, 12 in, 4 ohm, black anodised aluminium cone
- ⑬ **Trim rings x2**
printed, satin black or gunmetal, over each frame and its screws
- ⑭ **Midrange x1**
SB Acoustics Satori MR16P-8, 6.5 in papyrus cone
- ⑮ **Tweeter x1**
SB Acoustics Satori TW29DN-B, faceplate off, rear-mounted on the insert
- ⑯ **Amplifier x1**
Hypex FusionAmp FA253, DSP, 3 channels

earmilk floorstander | Exploded view and parts

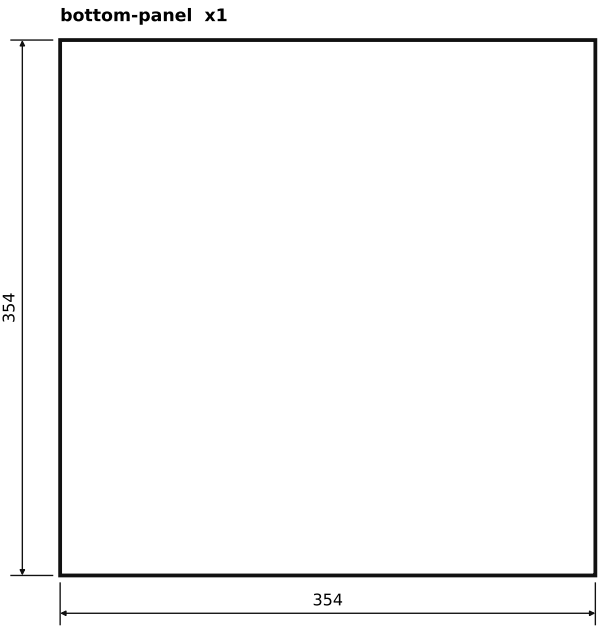
Isometric, not to scale: the parts pulled apart along the way they go together, and what each one is.

mm | issue 2026-10-09 | from fab/sheets.py (the CAD) | fits as dimensioned

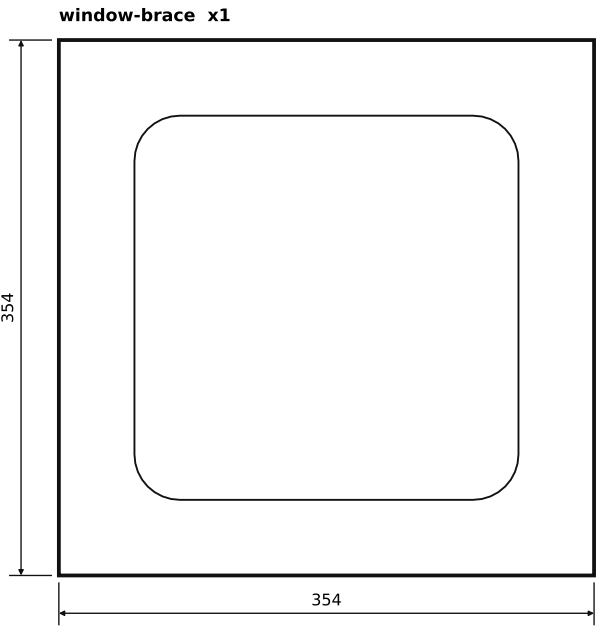
Tolerances: panels ± 0.5 , pocket depths ± 0.2 , holes $+0.2/-0$, prints ± 0.15 . Check bought parts first.



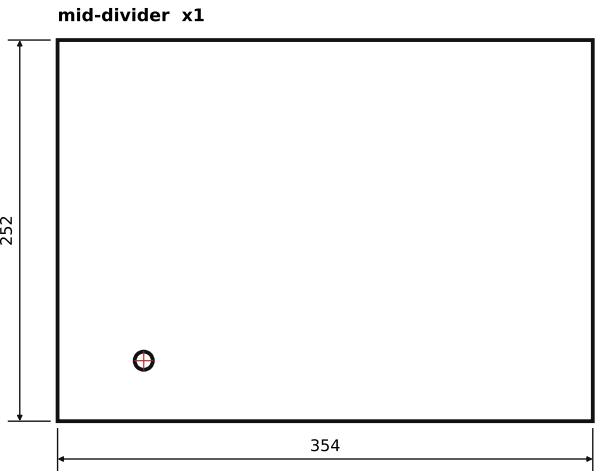
ø14 at (177, 215.5): cut inside
ø10 at (42, 197): drill ø10, 10 deep
ø10 at (312, 197): drill ø10, 10 deep
ø10 at (42, 312): drill ø10, 10 deep
ø10 at (312, 312): drill ø10, 10 deep
upper face up (front edge at the bottom of the drawing); dowels
register the gable block; the tweeter cable's hole gets a grommet



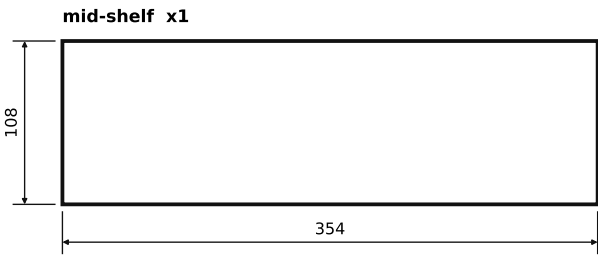
either face



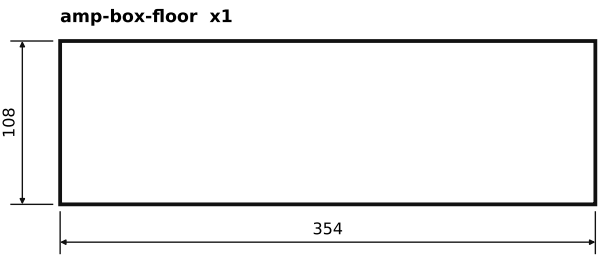
cut inside: 50 to 304 across, 50 to 304 up
sits at z 500 to 518, glued to all four walls



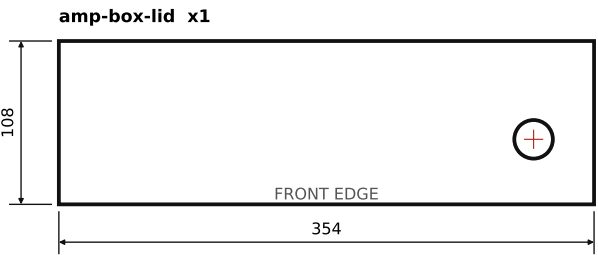
ø12 at (57, 40): cut inside
y 108 to 126, z 590 to 842; seal the wire hole after wiring



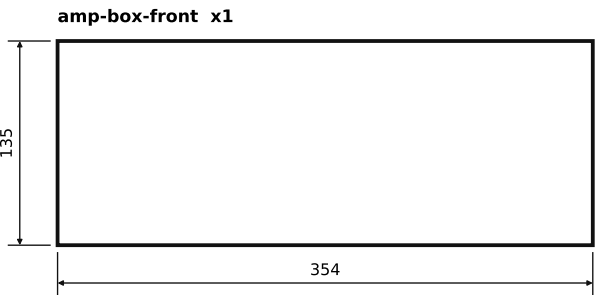
z 572 to 590, from the baffle back to under the divider



the amplifier box's floor, z 99.5 to 117.5, against the back



ø25.5 at (314, 43): cut inside
the amplifier box's lid, z 252.5 to 270.5, its front edge (the
bottom of the drawing) over the box's front; the gland's ø25.5
hole 314 from the left, 43 from the front edge; sealed after
wiring



the amplifier box's front, between floor and lid, 90 in front of
the back's inner face; glue and seal all round

earmilk floorstander | Panel details

Each panel seen from its outer (upper) face, from its lower-left corner as its DXF; holes from that corner. 18 mm Baltic birch.

mm | issue 2026-10-09 | from fab/sheets.py (the CAD) | fits as dimensioned

Tolerances: panels ± 0.5 , pocket depths ± 0.2 , holes $+0.2/-0$, prints ± 0.15 . Check bought parts first.